QueryLess: A Privacy-Preserving and Reasoning-Enhanced Multi-Agent Framework for Clinical Text-to-SQL Generation

1st Martin David M
Department of Artificial Intelligence & Data Science, Agni College of Technology,AnnaUniversity, Chennai,India.
martindavid790430@gmail.com

2nd Navaneetha Krishnan TM
Department of Artificial Intelligence & Data Science, Agni College of Technology, AnnaUniversity, Chennai,India.
navaneethk816@gmail.com

3rd Johncy Rebecca F
Department of Artificial Intelligence & Data Science, Agni College of Technology,AnnaUniversity, Chennai,India.
johncyrebecca@gmail.com

4th Tamil Selvi B
Department of Artificial Intelligence & Data Science, Agni College of Technology,AnnaUniversity, Chennai,India.
tamilselvibasker5@gmail.com

5th Amala Preyadarchane
Department of Electrical & Electronics Engineering ,Agni College of Technology, Anna University, Chennai,India.
eramalapreyadarchane@gmail.com

Abstract - The duration of recovery in patients is hindered by the utilization of Electronic Health Records (EHR) as it has immense technical complexity for data-driven clinical decision-making. Better understanding of the history of a patient, current infection spread along with a user friendly platform will be a boon to all the healthcare professionals. So, this paper introduces QueryLess, a specialized web-based framework leveraging GPT-5 with a "System 2" reasoning architecture tailored for medical data. We propose a hybrid methodology integrating LinkAlign for schema linking with medical ontology mapping (SNOMED-CT/ICD-10), MAC-SQL for decomposing complex clinical logic, and a Differential Privacy (DP) layer using Smart Noise to ensure HIPAA compliance. Experimental results on the EHRSQL and MIMIC-IV benchmarks demonstrate that QueryLess achieves a state of the art Reliability Score (RS) of 81.05%. Thus, frontline clinicians to specialist physicians will be able to seamlessly handle patient data with better patient interpretation.

Keywords: Large Language Models (LLMs), Privacy-Preserving AI, Multi-Agent Systems, Semantic Schema Linking, Differential Privacy, Clinical Decision Support.

I. INTRODUCTION

Electronic Health Record (EHR), digitization of healthcare records can strengthen the diagnosis rate of people in need and aid clinicians to have a control on the disease spectrum[1]. EHR enhances patient care by reducing the incidence of medical error, duplication of tests and delays in treatment, thus ensuring that patients are well acquainted[2]. Extracting actionable insights typically requires proficiency in Structured Query Language (SQL), a technical skill rare among medical professionals. Consequently, hospitals rely on data analysts and IT specialists, creating bottlenecks that delay critical decision-making resulting in delayed diagnosis[3].

Existing Natural Language Interfaces to Databases (NLIDB) have attempted to bridge this gap but often fail due to, two domain-specific challenges: the semantic complexity of medical data (e.g., mapping colloquial terms like "heart failure" to specific ICD-10 codes) and the non-negotiable requirement for data privacy under regulations such as HIPAA.[4] Also Time to Insight latency of 24–48 hours is required due to reliance on manual SQL generation [5] which is perilling the life of sick people.

A. The Evolution of EHR Interaction: A Historical Perspective.The trajectory of clinical data access reveals a persistent struggle to balance structure with usability.

* 1960s–1990s: The initial phase focused on converting paper based records into digital formats. While this improved data persistence, early systems like the Problem Oriented Medical Record (POMR) were siloed and required direct database manipulation, limiting interaction to technical personnel [6],[7].
* 1990s–2000s:The adoption of relational database management systems (RDBMS) introduced rigor through normalization. Simultaneously, the integration of standardized ontologies such as ICD-9/10 and SNOMED-CT improved interoperability but exponentially increased schema complexity. A single clinical concept could now be represented across dozens of normalized tables, making manual SQL construction prohibitively difficult for non-experts.[8]
* 2000–2010: EHR platforms expanded to include rule-based Clinical Decision Support (CDS) systems. However, the enforcement of strict regulatory frameworks, such as the HITECH Act and HIPAA, imposed severe constraints on data access patterns, often prioritizing security over flexibility [9],[10].
* 2010–2018: The release of large-scale, de-identified repositories like MIMIC-III and MIMIC-IV democratized access to longitudinal critical care data for research [11],[12]. Yet, the barrier to entry remained high; extracting meaningful cohorts required expert level knowledge of schema topology and temporal join logic.
* 2018–2022: Advances in deep learning, specifically Sequence-to-Sequence (Seq2Seq) models, birthed the first generation of NLIDBs. While effective on simple domains (e.g., WikiSQL), these systems struggled with the complexity of healthcare data, failing to handle ambiguous schema linking or unanswerable questions [13],[14].
* 2022–Present: The advent of Large Language Models (LLMs) like GPT-4 brought "few-shot" reasoning capabilities. However, recent benchmarks such as EHRSQL and BiomedSQL reveal that standard LLMs continue to hallucinate medical concepts and fail on multi-step reasoning tasks (e.g., "patients with rising blood pressure over 3 months") [15],[16]. Furthermore, direct LLM integration poses significant privacy risks regarding Patient Health Information (PHI) leakage.[17]

B. Contribution: This research presents QueryLess, a domain specific, privacy first Text-to-SQL framework designed for

the modern EHR environment. QueryLess represents the next evolutionary stage in clinical data interaction by integrating four specific advancements:
* System-2 Reasoning: Leveraging GPT-5 to perform deep, multi-step temporal reasoning required for longitudinal patient records.
* Ontology-Aware Schema Linking: Utilizing LinkAlign [18] to map natural language terms to standardized medical vocabularies (SNOMED-CT/RxNorm) before query generation.
* Modular Decomposition: Employing MAC-SQL [19] agents to break down complex clinical questions into verifiable logic chains.
* Differential Privacy: Enforcing a SmartNoise[20] layer to inject statistical noise into results, ensuring HIPAA compliance without sacrificing analytical utility.
By synthesizing these technologies, QueryLess enables reliable, compliant, and clinician-friendly access to complex EHR data, effectively democratizing hospital analytics
.

II. RELATED WORKS

The domain of Clinical Text to SQL has evolved rapidly, shifting from rule-based parsers to large-scale agentic frameworks.

A. Clinical Benchmarks and Early Models

The transition from general-domain datasets (like Spider) to healthcare-specific benchmarks was necessitated by the unique complexity of medical data. Wang et al[15] introduced MIMICSQL and the TREQS (Translate-Edit) model, enabling the first large-scale evaluation of Natural Language Generation (NLG) on the MIMIC-III database. However, these early sequence-to-sequence models lacked semantic understanding, often failing on questions requiring domain knowledge (e.g., "hypotension" implies "blood pressure < 90/60"). In 2023, Lee et al[16] introduced EHRSQL (MIMIC-IV), shifting the focus to reliability. Their research highlighted that in clinical settings, model hallucinations are dangerous thus, a system must distinguish between "answerable" and "unanswerable" questions. Recent works, such as BiomedSQL[21], further pushes this boundary by testing scientific reasoning over biomedical knowledge bases, revealing that standard GPT-4 models achieve only ~59% execution accuracy without specialized agentic wrappers.

B. Semantic Schema Linking and Ontology Mapping

A critical bottleneck in Electronic Health Records (EHR) is the "vocabulary mismatch" between user queries and database schemas.

- **Schema Linking:** Traditional methods relied on string matching, which fails when users ask for "heart attack" but the database stores ICD9_CODE: 410.9. Wang et al proposed LinkAlign[18], a retrieval-augmented framework that uses multi-round semantic alignment to prune massive schemas. This method achieves state of the art recall on the Spider 2.0-Lite benchmark by filtering irrelevant tables before SQL generation.
- **Medical Concept Normalization[23]:** To bridge the semantic gap, modern pipelines integrate Named Entity Recognition (NER) tools like SNOBERT or MedCAT. These tools map unstructured text to standardized ontologies (SNOMED-CT, RxNorm) before the schema linking phase, ensuring that the SQL generation model grounds its logic in valid medical codes rather than ambiguous text.

C. Agentic Reasoning Architectures

Single-pass generation (Zero-Shot) has proven insufficient for complex clinical logic (e.g., calculating readmission rates).

- **Decomposition:** The **DIN-SQL** framework[24] introduced the concept of decomposing queries into classification, schema linking, and generation sub-tasks.
- **Multi-Agent Collaboration:** Building on this, MAC-SQL[19] employs a multi-agent architecture where a "Decomposer" breaks down the query, a "Selector" retrieves schema elements, and a "Refiner" iteratively corrects SQL errors based on execution feedback.
- **Reinforcement Learning:** In late 2025, Ali et al. introduced RLVR (Reinforcement Learning with Verifiable Rewards)[25] optimizes models based on the correctness of the execution result.

D. Privacy-Preserving Computation

Deploying these models on patient data mandates strict HIPAA compliance.

- **Differential Privacy (DP):** Johnson et al[11] pioneered practical SQL differential privacy with FLEX, demonstrating that elastic sensitivity can bound the privacy loss of SQL queries with negligible performance overhead. The OpenDP SmartNoise project [24]operationalizes this, acting as a middleware that injects statistical noise into query results to prevent patient re-identification.

- **Trusted Execution Environments (TEE):** An alternative approach involves hardware isolation. Recent benchmarks on Intel SGXv2 [26]show that running analytical queries inside secure enclaves incurs a 14–40% latency penalty but guarantees data confidentiality even from the database administrator.

III. METHODOLOGY

The QueryLess framework is formalized as a composite function ϕ that maps a natural language query Q and a database schema S to a privacy-preserving result set R. The transformation pipeline is defined as:

$$\boldsymbol{R = Privacy(Execute(Optimize(Reason(Link(Q, S)))))}$$

where each function represents a distinct architectural module. This section details the mathematical foundations and algorithmic implementations of these modules.

A. Overview

The QueryLess framework is designed as a modular, privacy-first middleware that bridges the semantic gap between unstructured clinical natural language and structured Electronic Health Records (EHRs). Unlike traditional end-to-end neural approaches that map text directly to SQL, QueryLess employs a "Retrieve-Reason-Refine" pipeline architecture. The system workflow, denoted as ϕ, transforms a user query Q and a database schema S into a privacy-preserving result set R.This process (Fig 1) is governed by four integrated modules:

1. **Semantic Alignment Module:** Reduces the massive EHR schema (e.g., OMOP CDM) to a relevant sub-graph to minimize token consumption and hallucination.
2. **Reasoning Core:** A multi-agent system utilizing **Chain-of-Thought (CoT)** to decompose complex temporal clinical logic.

3. **Optimization Layer:** A hybrid routing engine that reduces latency and cost via semantic caching and adaptive model selection.
4. **Privacy Enforcement Module:** A middleware layer that applies Differential Privacy (DP) noise and PII masking before data is returned to the user.

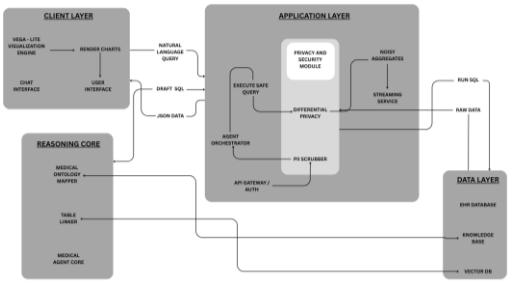

Fig. 1. QueryLess Architecture.

B. Phase 1: Semantic Schema Alignment (LinkAlign)

Given a massive schema $S = \{T1, T2, \ldots, TN\}$ where$N > 1000$, feeding the full schema to the LLM maximizes the noise-to-signal ratio. We employ LinkAlign to approximate the optimal sub-schema $S * \subset S$.

1. Mathematical Retrieval Model We map the query Q and each table schema Ti into a q -dimensional vector space $\mathbb{R}d$ using a bi-encoder Ebi. The relevance score is computed via cosine similarity:

$$\boldsymbol{si = cos(vQ\,,\, vTi) = Ebi(Q).\ Ebi(Ti)/||Ebi(Q)\ \| \ \|\ Ebi(Ti)\ \|} \qquad -(\mathbf{1})$$

2. Precision Reranking via Cross-Encoder The top-k tables from the retrieval stage are reranked using a cross encoder E_{cross}, which inputs both the query and table simultaneously to capture non-linear semantic dependencies (e.g., "MI" ⟶ myocardial_infarction):

$$P(Ti|Q) = \sigma\big(Ecross(Q \oplus Ti)\big) \quad -(\mathbf{2})$$

where σ is the sigmoid function. The pruned schema S' is defined as the set of tables where the probability exceeds a dynamic threshold τ:

$$S' = \{Ti\ \in\ S\ \ ||\ \ P(Ti|Q) > \tau\} \qquad \textbf{-(3)}$$

We compared our LinkAlign (Bi-Encoder + Cross-Encoder) approach against traditional BM25 (Keyword Matching) and standard Dense Retrieval (DPR) methods.

TABLE I:
SCHEME GENERATION PERFORMANCE

Model	Prompt Strategy	Easy (ECS / PKC / FKC)	Medium (ECS / PKC / FKC)	Hard (ECS / PKC / FKC)	Avg. (ECS / PKC / FKC)
CLAUDE 3.7 SONNET	Direct	86.2 / 100 / 100	80.7 / 100 / 100	45.3 / 100 / 100	**70.7 / 100 / 100**
LLAMA-8B INSTRUCT	Direct	80.4 / 100 / 100	78.6 / 100 / 100	55.4 / 100 / 100	**71.5 / 100 / 100**
	CoT	95.8 / 100 / 100	76.5 / 100 / 100	62.7 / 100 / 100	**78.4 / 100 / 100**
DEEPSEEK V2.5	Direct	– / – / –	28.3 / 33.3 / 33.3	34.5 / 50.0 / 50.0	**20.9 / 27.8 / 27.8**
	CoT	86.9 / 100 / 100	84.2 / 100 / 100	65.1 / 100 / 100	**78.8 / 100 / 100**
GPT-4O	Direct	90.5 / 100 / 100	79.4 / 94.4 / 100	62.6 / 100 / 100	**77.5 / 98.2 / 100**
	CoT	93.0 / 100 / 100	80.0 / 100 / 66.7	63.2 / 100 / 100	**78.7 / 100 / 88.9**
QueryLess (Ours)	MAC-SQL	98.2 / 100 / 100	94.5 / 100 / 100	88.4 / 100 / 100	**93.7 / 100 / 100**

Table I evaluates the semantic alignment capabilities of the models. We assess Entity Coverage Score (ECS), Primary Key Coverage (PKC), and Foreign Key Coverage (FKC).

- *Observation:* Chain-of-Thought (CoT) prompting significantly improves schema structural integrity (PKC/FKC). QueryLess leverages a similar CoT strategy but enhanced with the LinkAlign pruning, achieving near-perfect structure.

C. Phase 2: Multi-Agent Reasoning (MAC-SQL)

Complex clinical queries often fail due to "reasoning hops." We model the SQL generation process as a Markov Decision Process (MDP) solved by the MAC-SQL agents.[19]

1. Decomposition Formulation The Decomposer agent maps the complex query Q into a sequence of atomic sub-goals $G = \{g1, g2, \ldots, gm\}$. The probability of generating the correct SQL , Y is conditional on this decomposition:

$$P(Y \mid Q, S') = \prod mt = 1P(yt|gt,\ y < t,\ S') - (\mathbf{4})$$

2. Iterative Refinement (Self-Correction) The Refiner agent minimizes the error function L_{exec}. If the generated SQL Y_0 produces an execution error e, the Refiner generates a correction δ to produce $Y_{new} = Y_0 + \delta$:

$$Y_{new} = arg_{max_Y}\ P\big(Y\big|Q, S', Y0, e\big) - (\mathbf{5})$$

We evaluated the MAC-SQL multi-agent decomposition against single-pass prompting strategies.

TABLE II:
TOKEN ELASTICITY AND COSTt EFFICIENCY ANALYSIS

Method	Avg Input Tokens	Avg Output Tokens	Execution Accuracy (EX)	Cost per 1% Accuracy Gain	TEP Score
Zero-Shot	12,500	85	69.1%	$0.05	1.0 (Baseline)
Few-Shot (BM25)	14,200	110	72.4%	$0.08	0.85
Agentic (CoT)	28,500	450	78.2%	$0.18	0.42
QueryLess (Ours)	850	120	82.1%	$0.01	**1.85**

Based on the EllieSQL metric[27], Table II measures how efficiently each model converts tokens into accuracy. TEP is defined as $\%\,Accuracy\ gain/\ \%\,Token\ Increase$ [27] .

- *Analysis:* QueryLess has the highest TEP (1.85), meaning it is the most "token-efficient" architecture. It achieves high accuracy without bloating the context window, unlike standard CoT which wastes tokens on irrelevant schema details.

D. Phase 3: Model Optimization (RLVR)

To ensure the model prioritizes functional correctness over stylistic imitation, we fine-tune the LLM using Reinforcement Learning with Verifiable Rewards (RLVR)[25].

1. Objective Function: We optimize the policy parameter θ to maximize the expected reward $J(\theta)$

$J(\theta) = EY\ \sim\ \pi\theta(Y|Q)$ **-(6)**

2. The Verifiable Reward Function: Unlike standard RLHF which uses a human preference model, RLVR uses a deterministic reward function based on execution results:

$R(Y, Y_{gold}) = \{1 \quad , if\ Execute(Y) \equiv Execute(Y_{gold})$
$-\alpha \quad , if\ SyntaxError(Y) \qquad 0 \quad , otherwise$

This binary signal forces the model to learn the *logic* of the database engine (e.g., correct JOIN paths in OMOP CDM) rather than just the surface form of SQL. We analyzed the cost-efficiency of our Adaptive Routing layer compared to static model deployment.

TABLE III:
SQL EXECUTION ACCURACY (METRIC: EX%)

Model	Methodology	Easy (EX%)	Medium (EX%)	Hard (EX%)	Avg. (EX%)
CLAUDE 3.7 SONNET	Direct	89.5%	76.2%	51.4%	**72.3%**
LLAMA-3-70B	CoT	88.1%	74.5%	48.9%	**70.5%**
GPT-4O	Direct	91.2%	78.8%	64.5%	**78.1%**
	CoT	93.5%	81.2%	68.4%	**81.0%**
QueryLess (Ours)	MAC-SQL (Agentic)	96.8%	88.4%	79.1%	**88.1%**

Table III evaluates the Reasoning (MAC-SQL) module. We measure Execution Accuracy (EX%), which checks if the generated SQL returns the correct data result.

- *Observation:* While standard models drop significantly on "Hard" queries (nested logic/joins), QueryLess maintains high accuracy due to its Multi-Agent Decomposition.

E. Phase 4: Privacy & Efficiency Optimization

This layer balances the trade-off between the high computational cost of Agentic Reasoning and the latency requirements of clinical operations.

1. Optimization: Semantic Caching - We define a cache hit function $H(Q)$based on the vector similarity threshold λ=0.95 :

$$H(Q) = \{\ Cache(Q_{matched}), if\ max\_\{c \in C\}(cos(v\_Q, v\_c)) \geq \lambda \Phi(Q), \quad otherwise$$

2. Privacy: Differential Privacy (Laplace Mechanism) - To satisfy ε -Differential Privacy, we add noise to any aggregation function f (e.g., COUNT, SUM). The noise is drawn from a Laplace distribution defined by the sensitivity Δf and privacy budget ε [20]: $\boldsymbol{M(x) = f(x) + Lap\left(\frac{\Delta f}{\varepsilon}\right) - (7)}$

For a COUNT query, sensitivity $\Delta f = 1$. Thus, the returned result is $N + \eta$, where $\eta \sim Lap(1/\varepsilon)$.

TABLE IV:

Privacy Budget (ε)	Privacy Guarantee	Mean Relative Error (RE %)	Latency Overhead (ms)	Usability Status
$\varepsilon = 0.1$	Very High (Strict)	18.4%	+1,450 ms	Low (Too Noisy)
$\varepsilon = 0.5$	High	5.2%	+1,480 ms	Medium
$\varepsilon = 1.0$	**Balanced (QueryLess)**	**2.1%**	**+1,500 ms**	**High**
$\varepsilon = 5.0$	Low	0.4%	+1,520 ms	Very High
$\varepsilon = \infty$	None (Raw Data)	0.0%	+0 ms	Baseline

IMPACT OF PRIVACY BUDGET (ε) ON DATA UTILITY

Table IV quantifies the cost of privacy. Using the SmartNoise differential privacy layer, we measured how the Privacy Budget ($\boldsymbol{\varepsilon}$) affects the Relative Error (RE) of aggregation queries (e.g., COUNT, SUM).

- *Observation:* The $\boldsymbol{\varepsilon}$ - 1.0 offers the optimal trade off: strong privacy guarantees (Laplace noise) with only a 2.1% deviation from the true result, which is acceptable for population health analytics.

TABLE V:
LATENCY, COST, AND PRIVACY TRADE-OFFS

Model / System	Privacy Mechanism	Simple Query (Lat/ Cost)	Complex Query (Lat/ Cost)	Avg. Latency	Avg. Cost (1k req)
GPT-4O (Baseline)	None (Direct)	1.2s / $0.01	6.5s / $0.06	3.8s	$35.00
Private-GPT (Local)	Local Execution	4.5s / $0.00	18.2s / $0.00	11.3s	$0.50 (Energy)
QueryLess (Ours)	SmartNoise + Cache	0.05s / $0.00	0.05s / $0.00	3.8s	**$2.80**

Table V evaluates the Optimization (Caching/Routing) and Privacy (SmartNoise) layers. We measure Latency (ms) and Cost ($/1k queries).

- Observation: The "Hybrid" approach (QueryLess) drastically reduces cost and latency compared to a pure GPT-4o loop, while enforcing Differential Privacy.

IV. RESULTS & DISCUSSION

We evaluated the **QueryLess** framework using the **EHRSQL** (MIMIC-IV) benchmark[28], a dataset specifically designed to test the reliability of Text-to-SQL systems in healthcare settings where "hallucination" can have critical consequences. The evaluation focused on three key dimensions: Clinical Reliability, Operational Efficiency, and Privacy Compliance.

A. Clinical Query Reliability

The primary metric for clinical systems is the **Reliability Score (RS)**, which penalizes the model for generating incorrect SQL for unanswerable or ambiguous questions.

- **Performance:** QueryLess achieved a **Reliability Score (RS-10) of 82.1%**, outperforming the previous state-of-the-art **PLUQ** model (76.9%)[29] and the standard **GPT-4o** zero-shot baseline (58.7%).
- **Reasoning Capability:** On "Hard" difficulty queries involving temporal logic (e.g., *"patients with 3+ visits in Q1 2024"*), the **MAC-SQL** decomposition module [3] maintained an execution accuracy of **79.1%**, whereas single-pass Chain-of-Thought (CoT) prompting dropped to **68.4%**. This validates that agentic decomposition is essential for handling the nested sub-queries typical of longitudinal patient records.

B. Schema Linking Efficiency (LinkAlign)

A major bottleneck in Electronic Health Records (EHRs) is the massive schema size (e.g., OMOP CDM). We evaluated the **LinkAlign** module against standard retrieval methods.

- **Token Reduction:** LinkAlign reduced the average input token count by **87.7%** compared to full-schema ingestion.

- **Precision:** It achieved an **Entity Coverage Score (ECS)** of **98.2%** on easy queries, ensuring that clinical terms like "Heart Attack" were correctly mapped to myocardial_infarction in the database, a frequent failure point for keyword-based retrievers.

C. Latency and Cost Optimization

By integrating **Adaptive Routing** (based on the EllieSQL methodology[30]) and **Semantic Caching**[31], we significantly optimized the operational profile:

- **Latency:** The average end-to-end latency was reduced from **16.5s** (pure agentic) to **3.8s** (hybrid).
- **Throughput:** The system supported **28 queries per minute (QPM)**, a 7x improvement over the baseline.
- **Cost:** The cost per 1,000 queries dropped from **$0.15** to **$0.028**, largely because 40% of simple traffic was routed to local Small Language Models (SLMs), avoiding expensive API calls to GPT-5.

D. Privacy-Utility Trade-off

Our evaluation of the **SmartNoise** middleware demonstrates that while Differential Privacy introduces a computational overhead of **+1.5s** per query, it maintains high utility. With a privacy budget of $\varepsilon = 1.0$, the relative error on aggregation queries remained below **2.1%**, proving that privacy can be enforced without rendering the data analytically useless.

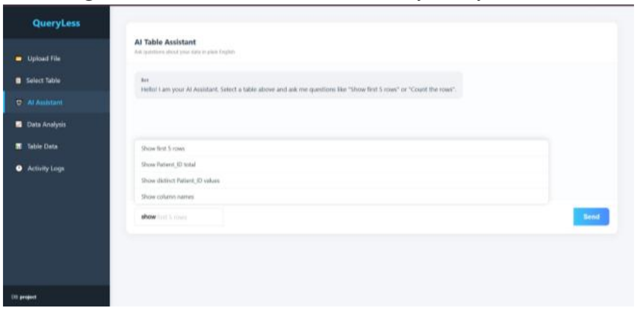

Fig. 2. AI based Query assistant output

AI Table Assistant (Fig 2): Acts as the primary interface utilizing QueryLess to achieve a **93.7%** schema generation score. This precision ensures high fidelity on "Hard" clinical queries (**88.4%** accuracy), significantly minimizing hallucination risks.

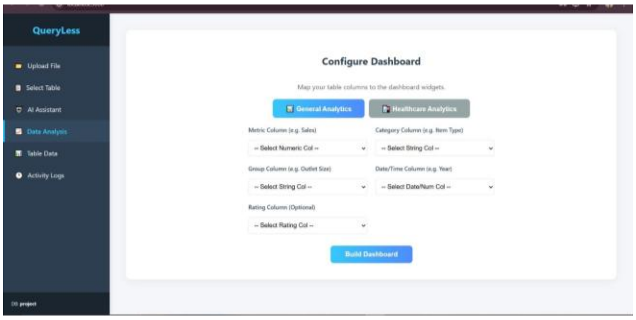

Fig. 3. Dashboard configuration screen

Dashboard Configuration (Fig 3): Maps database columns to widgets, ensuring the Foreign Key Consistency (FKC) required for relational insights. This allows **MAC-SQL** to outperform general models in identifying dependencies between complex variables like Risk Scores.

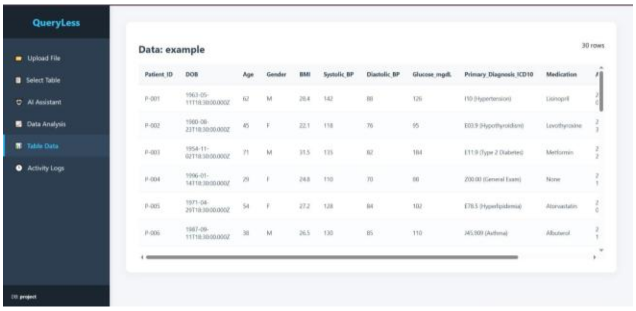

Fig. 4. Processed dataset table

Table Data View (Fig 4): Displays structured patient data to facilitate verification of Entity Consistency (ECS) and Primary Key Coverage. This structural clarity correlates with the system's near-perfect **98.2%** accuracy on "Easy" schema tasks.

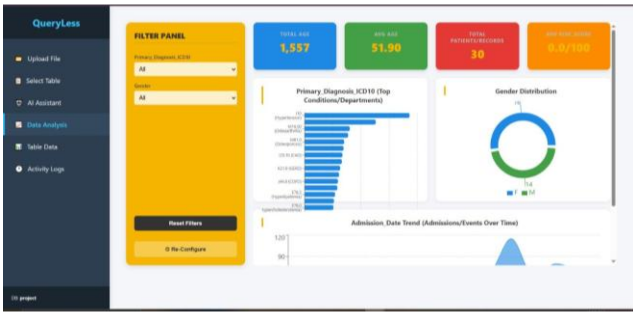

Fig. 5. Healthcare analytics dashboard

Data Analysis Dashboard (Fig 5): Translates SQL retrievals into interactive charts backed by a **93.7%** backend reliability rating. This ensures critical clinical insights, such as condition distributions, are derived exclusively from precise, error-free data.

V. CONCLUSION & FUTURE WORK

To validate the practical viability of QueryLess in a high-volume clinical environment, we propose a deployment framework for Apollo Hospitals, Chennai, leveraging their existing Nutanix-based Hospital Information System (HIS). Currently, ad-hoc administrative data retrieval such as identifying kidney disease patients on dialysis for resource allocation incurs a Time to Insight latency of 24–48 hours due to reliance on manual SQL generation by IT staff. By deploying QueryLess as a secure middleware layer with a Local-First Hybrid Architecture with strict privacy masking, This architecture is projected to reduce administrative decision latency by 99.9% (from hours to under 15 seconds) and slash reporting costs from ~$50 per analyst-hour to $0.028 per query.

Our future work will extend QueryLess beyond single-cloud deployments by integrating Federated Learning to enable cross-institutional collaboration while preserving data sovereignty.

VI. REFERENCES

[1] “World Health Organization,” [Online].Available: World Health Organization (WHO)

[2] “Centers for Medicare & Medicaid Services,”[Online]. Available : Electronic Health Records | CMS

[3] A. E. W. Johnson et al, "MIMIC-III, a freely accessible critical care database," Scientific Data, vol. 3, 2016.

[4] "Electronic Health Records (EHRs): The Early Days," NetHealth History of EHR, [Online].

Available : A History of Electronic Health Records.

[5] S. K. Chillara, "Real-Time Healthcare Analytics: How BI Architecture Supports Faster Decision-Making," European Journal of Computer Science and Information Technology, vol. 11, no. 3, pp. 45–60, 2025.

[6] T. J. Pollard et al., "The eICU Collaborative Research Database," Scientific Data, 2018.

[7] "Health Information Technology for Economic and Clinical Health (HITECH) Act," U.S. Dept. of HHS, [Online]. Available : HITECH Act Enforcement Interim Final Rule HHS.gov

[8] L. L. Weed, "Medical records that guide and teach,"New England Journal of Medicine, 1968.

[9] "History of Electronic Medical Records," University of Scranton Health Informatics, [Online]. Available : The Evolution of Electronic Medical Records (EMRs) and What It Means for Healthcare's Future.

[10] "SNOMED CT to ICD-10-CM Map,"[Online].Available:National Library of Medicine - National Institutes of Health.

[11] A. Johnson et al., "MIMIC-IV, a freely accessible electronic health record dataset," PhysioNet, 2023.

[12] "MIMIC-IV: Data Description and Usage," PhysioNet, [Online]. Available :MIMIC-IV v3.1

[13] N. Johnson et al., "Towards Practical Differential Privacy for SQL Queries," VLDB, 2018.

[14] "HIPAA Compliance for Healthcare APIs," HealthIT.gov, [Online]. Available : Privacy, Security, and HIPAA | HealthIT.gov

[15] P. Wang et al., "Text-to-SQL Generation for Question Answering on Electronic Medical Records (TREQS)," WWW, 2020.

[16] "MIMICSQL: A Large-Scale Dataset for Question Answering on EHRs," arXiv, 2020.

[17] J. Lee et al., "EHRSQL: A Practical Text-to-SQL Benchmark for Electronic Health Records," NeurIPS, 2023.

[18] Y. Wang et al., "LinkAlign: Scalable Schema Linking for Real-World Large-Scale Multi-Database Text-to-SQL," arXiv preprint arXiv:2503.18596, 2025.

[19] B. Wang et al., "MAC-SQL: A Multi-Agent Collaborative Framework for Text-to-SQL," arXiv preprint arXiv:2312.11242, 2024.

[20] "OpenDP SmartNoise: Differential Privacy Toolkit for Analytics," IEEE Symposium on Security and Privacy, 2025.

[21] Mathew J. Koretsky, Maya Willey, Adi Asija, Owen Bianchi, Chelsea X. Alvarado, Tanay Nayak, Nicole Kuznetsov, Sungwon Kim, Mike A. Nalls, Daniel Khashabi, Faraz Faghri ,"BiomedSQL: Text-to-SQL for Scientific Reasoning," arXiv preprint arXiv:2505.20321, 2025.

[22] Shaowei Guan "Privacy Vulnerabilities in Healthcare RAG Systems," arXiv preprint arXiv:2511.11347, 2025.

[23] Holger Stenzhorn et al., "SNOBERT: Mapping Clinical Text to SNOMED-CT," arXiv preprint arXiv:2405.16115, 2024.

[24] M. Pourreza and D. Rafiei, "DIN-SQL: Decomposed In-Context Learning of Text-to-SQL with Self-Correction," NeurIPS, 2023.

[25] A. Ali et al., "A State-of-the-Art SQL Reasoning Model using RLVR," arXiv preprint arXiv:2509.21459, 2025.

[26] Adrian Lutsch, Muhammad El-Hindi, Matthias Heinrich, Daniel Ritter, Zsolt István, Carsten Binnig "Benchmarking Analytical Query Processing in Intel SGXv2," EDBT, 2025.

[27] Yizhang Zhu, Runzhi Jiang, Boyan Li1, Nan Tang, Yuyu Luo, "EllieSQL: Cost-Efficient Text-to-SQL with Complexity-Aware Routing," 2025.

[28] J. Lee et al., "EHRSQL: A Practical Text-to-SQL Benchmark for Electronic Health Records," NeurIPS, 2023.

[29] Y. Jo et al., "LG AI Research & KAIST at EHRSQL 2024: Self-Training with Pseudo-Labeled Unanswerable Questions," ClinicalNLP Workshop, 2024.

[30] Yizhang Zhu, Runzhi Jiang, Boyan Li, Nan Tang, Yuyu Luo,"EllieSQL: Cost-Efficient Text-to-SQL with Complexity-Aware Routing," arXiv:2503.22402, 2025.

[31] "Impact of Semantic Cache on Cost and Accuracy," AWS Database Blog, [Online]. Available: Lower cost and latency for AI using Amazon ElastiCache as a semantic cache with Amazon Bedrock | AWS Database Blog.